AgriConnect - Use Case Model

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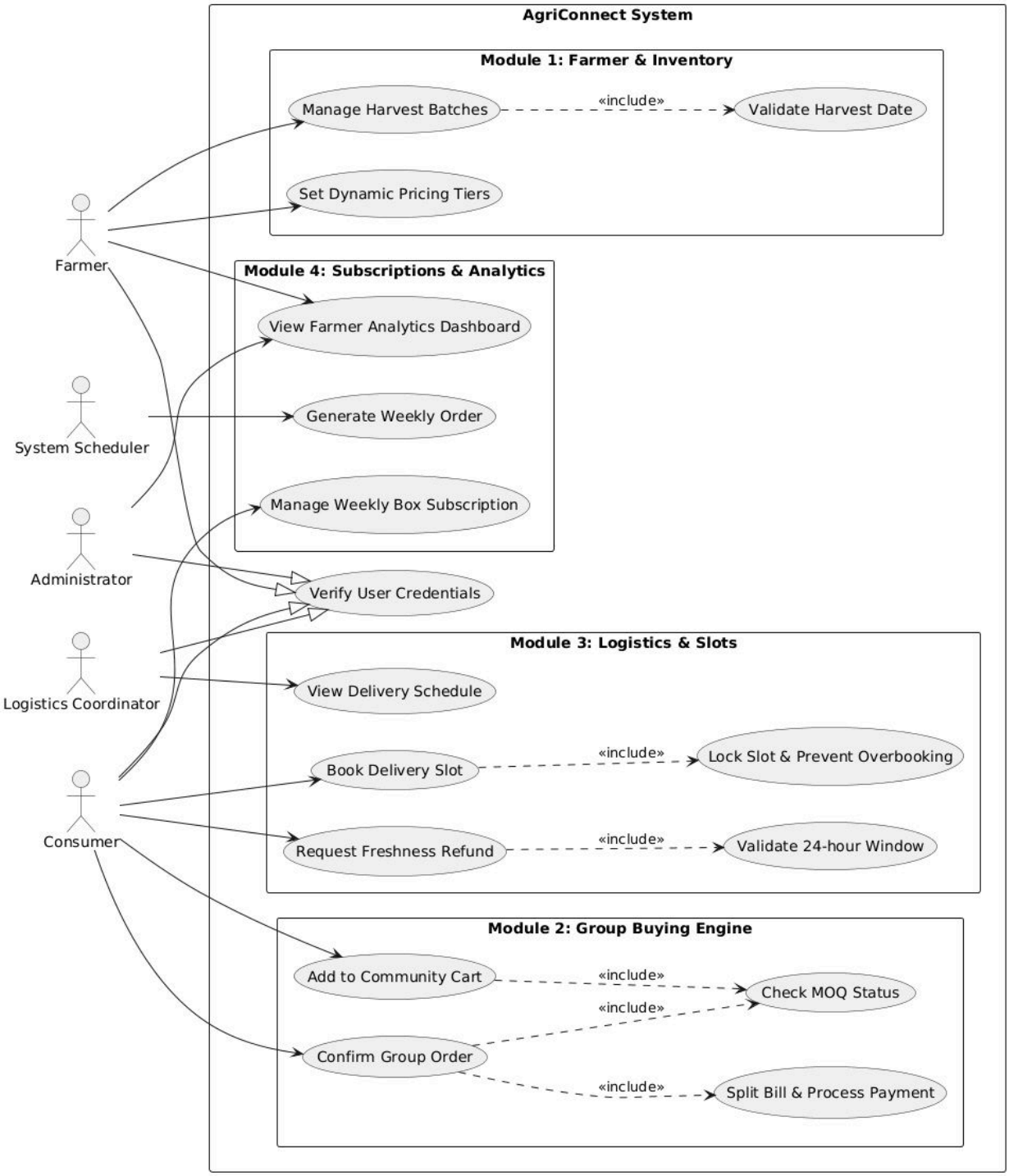

Actor Roles & Responsibilities

- **Farmer:** Manages the supply side by creating **Harvest Batches**, setting **Dynamic Pricing Tiers**, and monitoring business health via the **Analytics Dashboard**.
- **Consumer:** Drives the demand side by joining **Community Carts**, booking **Delivery Slots**, managing **Weekly Subscriptions**, and requesting refunds for damaged goods.
- **Administrator & Logistics Coordinator:** The Admin oversees user verification and analytics, while the Coordinator manages delivery schedules and slot availability.
- **System Scheduler:** An automated actor that triggers the background `generateWeeklyOrder()` function for subscriptions.

Module Breakdown

1. Farmer & Inventory Management

- **Core Action:** Farmers instantiate `HarvestBatch` objects.
- **Mandatory Logic:** Every batch creation **includes** a `validateDate()` check to ensure the harvest date is not in the past.

2. Group Buying Engine

- **Community Demand:** Users add items to a cart, which aggregates into a `GroupOrder`.
- **Validation:** Confirming an order **includes** checking the Minimum Order Quantity (MOQ). Only then does it trigger the **Split Bill** logic to calculate discounted shares.

3. Logistics & Slot Management

- **Freshness Control:** Includes a mandatory **Lock Slot** function during checkout to prevent overbooking.
- **Returns:** Refund requests **include** a `validateFreshness()` check to ensure they are filed within 24 hours of delivery.

4. Subscriptions & Analytics

- **Automation:** The system automatically generates weekly orders for active subscribers.
- **Insights:** Farmers and Admins access `AnalyticsReport` data to track revenue and wastage percentages.